VINAY MUPPALLA

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Portfolio | Research evidence | github.com/vinaychowdary9946-blip

SOFTWARE ENGINEERING

Computer Information Systems graduate student focused on backend reliability, data integrity and testable software. Builds Python systems with documented failure experiments and inspectable evidence.

TECHNICAL SKILLS

Python, Java, SQL, SQLite, Git; object-oriented programming, data structures, algorithms, HTTP APIs, unit and integration testing, debugging, SDLC and documentation. Additional tools: Excel, SAP GUI and ERP systems.

INDEPENDENT SYSTEMS STUDY

Effect Boundary Lab | 2026: Investigates retries after a remote service forgets its duplicate-protection receipt. Python, HTTP, separate SQLite stores and subprocess fault injection; raw evidence and a reproducible protocol are publicly available.

Published artifact reports 1,152 controlled trials. Guarded retry had zero excess effects and zero false confirmations across 288 trials, while quarantining 72 operations. These deterministic synthetic results are scoped to the experimental fixture.

SELECTED SOFTWARE PROJECTS

Resilient Job Queue: Durable background processing with leases and retries; a documented process-crash lab examines duplicate side effects and recovery. Python backend, local API and interactive browser replay.

Dependency Planner: Graph-based task scheduling with cycle detection, critical-path analysis and slack calculations; interactive demonstration and Python implementation.

Inventory Ledger: Event-history-based stock reconstruction with duplicate-command handling, overselling checks and version-conflict detection.

EDUCATION

M.S., Computer Information Systems - Rivier University | GPA: 3.79/4.0

B.Tech, Computer Science & Engineering - KITS, Guntur | CGPA: 7.2/10

ACADEMIC PROJECTS

SMART START - IoT Security System | Team Lead: Authentication-enabled vehicle ignition, sensor integration, safety mechanisms, debugging and system validation.

Business Acquisition Case Study: Supply-chain and operations analysis, workflow improvements and documented recommendations.